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NeuroState: A Comparative Framework for Electroencephalography-Based Meditative State Classification

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Abstract: Accurate identification of meditative states using electroencephalography (EEG) is crucial for objective neural assessment and brain-computer interface applications. However, reliable classification remains challenging because of signal nonstationarity, artifacts, inter-individual variability, and differences across recording sessions. To address these challenges, NeuroState, a dual-path classification framework integrating time-frequency analysis, conventional machine learning, and attention-enhanced deep learning, was developed for automated classification of meditation-related cognitive states. Initially, EEG signals were transformed into time-frequency representations using the Morlet continuous wavelet transform (CWT), allowing temporal variations in spectral activity to be characterized. Two complementary classification pathways were subsequently implemented. In the conventional machine-learning pathway, wavelet-derived scalograms were vectorized, and dimensionality was reduced using principal component analysis (PCA) before classification with support vector machine and random forest classifiers. In the deep-learning pathway, the two-dimensional scalograms were directly processed using a wide squeeze-and-excitation residual network, in which residual feature extraction and channel-wise attention were integrated to facilitate the adaptive learning of discriminative time-frequency representations. The framework was evaluated using the publicly available OpenNeuro ds001787 dataset, comprising EEG recordings from 24 participants, including 12 experienced meditators and 12 individuals with limited meditation experience. The wide squeeze-and-excitation residual network achieved a test accuracy of 98.29%, outperforming the support vector machine and random forest classifiers, which achieved accuracies of 88.59% and 81.65%, respectively. These findings demonstrate the superior classification performance of attention-enhanced deep learning compared with the evaluated conventional machine-learning approaches. These findings suggest that attention-enhanced deep learning applied to structured EEG time-frequency representations may provide more accurate discrimination of meditation-related cognitive states than conventional machine-learning approaches based on PCA-reduced features. NeuroState thus provides a promising computational framework for EEG-based meditative state classification, with potential applications in meditation monitoring, neurofeedback, and brain-computer interface systems.

Keywords: Eelectroencephalography; Continuous wavelet transform; Deep learning; Principal component analysis; Support vector machine; Random forest

1 Introduction

Meditation has long been recognized as a practice capable of influencing attention, awareness, and other aspects of cognitive functioning. With the development of modern neuroimaging and electrophysiological techniques, it has become possible to investigate the neural activity associated with different meditative conditions in an objective manner. Among the available techniques, electroencephalography (EEG) is particularly suitable for this purpose because it is non-invasive, relatively accessible, and provides excellent temporal resolution, allowing rapid variations in brain activity to be captured with millisecond precision [1]. Previous investigations have reported characteristic changes in EEG activity during meditation, particularly within the theta (4–8 Hz) and alpha (8–13 Hz) frequency ranges. Such frequency-dependent alterations suggest that meditative states possess measurable neural characteristics that can potentially be identified using computational classification techniques.

Despite these observations, developing a robust automated system for detecting meditation-related brain states remains challenging. One important limitation is the susceptibility of EEG recordings to physiological and environmental artifacts, including ocular activity, muscle movements, and other sources of interference [2]. In addition, substantial inter-individual differences in brain activity can lead to significant variations in EEG patterns among participants, making it difficult for a classifier trained on one group of subjects to generalize effectively to others. Another limitation arises from conventional signal processing approaches, in which complex EEG information is frequently summarized through a relatively small set of handcrafted statistical or spectral features. Such transformations may discard important relationships between temporal, spectral, and spatial characteristics of the original signal . These limitations motivate the use of learning architectures capable of extracting higher-level representations directly from structured EEG data.

To investigate these issues systematically, this study presents NeuroState, a unified framework for evaluating conventional machine learning and deep learning strategies for meditation state classification. A common EEG preprocessing pipeline is employed to ensure that both approaches operate on comparable input representations. Following preprocessing, the framework is divided into two complementary classification branches. The first branch represents a conventional machine learning strategy in which dimensionality reduction using principal component analysis (PCA) is followed by classification with the support vector machine and random forest algorithms [3, 4]. The second branch employs a wide squeeze-and-excitation residual network designed to learn discriminative patterns directly from the structured EEG representations.

The comparative design of NeuroState enables the contribution of the underlying model architecture to be examined without introducing differences in the initial data processing procedure. In particular, the deep learning branch incorporates residual learning and channel attention mechanisms to improve the extraction and weighting of informative features. Building upon earlier investigations of EEG characteristics associated with meditation, including the work of Usgaonkar et al. [2], this study explores whether attention-enhanced residual learning can provide a more effective representation of meditation-related neural activity than conventional classification techniques. The resulting framework therefore provides a systematic basis for assessing the potential of deep neural architectures for automated recognition of cognitive states from EEG recordings [5, 6].

The main contribution of this study is the development of NeuroState, a comparative framework that combines two-dimensional EEG power representations with a wide squeeze-and-excitation residual network for meditation experience classification. Unlike conventional approaches that flatten EEG features into numerical vectors for classification, the proposed deep learning branch preserves the channel–frequency structure of the 19×4 power maps derived from Morlet wavelet decomposition [7]. The framework further integrates progressively widened residual blocks with squeeze-and-excitation attention to learn hierarchical feature representations and adaptively recalibrate feature channel responses, potentially improving the discrimination of informative EEG patterns. Unlike the convolutional neural network approach reported in earlier meditation classification research, the proposed framework combines wider residual feature extraction and squeeze-and-excitation-based channel attention and evaluates this design alongside PCA-based support vector machine and random forest classifiers using a common preprocessing pipeline [8]. This comparative design enables an assessment of the benefits of structure-preserving representation learning and attention-enhanced residual classification for distinguishing experienced meditators from novice participants.

2 Literature Review

Research on EEG-based cognitive state recognition has progressively moved from manually designed signal descriptors toward automated representation learning. Early classification studies commonly relied on handcrafted spectral characteristics, such as band power measurements and power spectral density, extracted from selected EEG electrodes. The resulting numerical feature vectors were then provided to conventional classifiers, including support vector machines and k-nearest neighbors. These approaches offer the advantages of relatively low computational complexity and straightforward interpretation. However, their effectiveness depends strongly on the quality and selection of the manually engineered features. Moreover, representing EEG activity through a limited set of predefined descriptors may not adequately characterize its nonlinear, dynamic, and nonstationary nature.

Tree-based ensemble learning provides another direction for improving the robustness of EEG classification. Among these methods, the random forest combines predictions from multiple decision trees constructed using randomized subsets of training observations and input features. This ensemble strategy generally provides greater resistance to noisy and high-dimensional data than an individual decision tree. Nevertheless, random forests and similar conventional classifiers remain dependent on externally defined feature representations and therefore do not inherently learn hierarchical representations directly from the raw or structurally organized EEG signal.

The development of standardized EEG processing frameworks has also contributed significantly to the reproducibility of neurophysiological studies. MNE-Python, for example, provides established implementations for several important stages of EEG processing, including bandpass and notch filtering, common average referencing,

independent component analysis for artifact correction, and segmentation of continuous recordings into epochs. The availability of such standardized processing procedures helps minimize inconsistencies that can arise when preprocessing steps are implemented independently by different researchers.

A notable transition from handcrafted features to image-based representations was demonstrated by Usgaonkar et al. [2]. Their approach transformed EEG recordings from 19 selected channels into time-frequency images using Morlet wavelet analysis. Instead of supplying manually summarized spectral measurements to conventional classifiers, the resulting representations were processed using a convolutional neural network, enabling the network to learn discriminative patterns from the time-frequency structure. The reported classification accuracies reached 99.30% for the theta band, 98.55% for the alpha band, and 91.35% for the gamma band. These results surpassed the performance obtained from the classical methods examined in that study, including support vector machines, k-nearest neighbors, decision trees, and naive Bayes, as well as the evaluated deep architectures AlexNet and the Visual Geometry Group 19-layer (VGG-19) network. Importantly, evaluation using an additional independent dataset provided evidence that the proposed approach was not exclusively dependent on the characteristics of a single participant cohort.

The emergence of deep neural networks has subsequently broadened the possibilities for EEG representation learning. Schirrmeister et al. demonstrated that deep convolutional neural network architectures can learn EEG decoding features in an end-to-end framework, reducing the reliance on manually constructed descriptors. The introduction of residual learning further improved the training of deeper neural networks by providing shortcut connections that facilitate information propagation and gradient flow. In parallel, squeeze-and-excitation networks. Introduced a channel-attention mechanism that adaptively assigns greater importance to informative feature channels while reducing the contribution of less relevant ones. Such attention mechanisms are particularly relevant to EEG analysis because task-related information may not be uniformly distributed across all electrode channels or learned feature maps.

Building on these developments, NeuroState integrates several complementary concepts within a unified deep learning architecture. The proposed model combines residual connections, squeeze-and-excitation-based channel attention, and wider convolutional feature representations to enhance the extraction of discriminative information from EEG time-frequency data. Unlike conventional approaches that rely on manually selected feature vectors, the proposed framework retains the structured representation of the signal and allows the network to learn relevant spatial and spectral patterns automatically. The performance of this architecture is subsequently compared with classical machine learning baselines under consistent experimental conditions, enabling a direct assessment of the contribution provided by the proposed deep representation learning strategy.

3 Methodology

3.1 Dataset

The experimental evaluation in this study was conducted using the publicly accessible OpenNeuro dataset [9]. This meditation experiment contains 24 subjects. Subjects were meditating and were interupted about every 2 minutes to indicate their level of concentration and mind wandering. The dataset was collected at the Meditation Research Institute in Rishikesh, India, as part of an investigation into the neural characteristics associated with meditation practice. EEG recordings were obtained using a BioSemi ActiveTwo electrophysiological recording system, which provides multichannel acquisition suitable for investigating spatially distributed patterns of brain activity. A total of 64 scalp-mounted EEG electrodes were used during data collection, providing relatively dense spatial coverage of the cerebral cortex.

The EEG signals were initially acquired at a sampling frequency of 2048 Hz. Such a high acquisition rate allows the recorded signals to preserve fine temporal information and provides sufficient bandwidth for subsequent signal processing operations. For computationally efficient analysis, the recordings were subsequently downsampled to 256 Hz. This reduction in sampling frequency substantially reduced the computational and storage requirements while retaining the frequency components relevant to the cognitive state analysis performed in this work.

The dataset contains EEG recordings from 24 participants, who were categorized into two groups according to their level of meditation experience. The first group consisted of 12 experienced practitioners, identified as subjects sub-001 through sub-012. These participants had experience with Himalayan Yoga meditation and represented the experienced meditator category used in the classification experiments. On average, members of this group reported approximately two hours of meditation practice per day for at least one year. The mean age of the expert group was 39.3 ± 12.0 years, indicating variation in age while maintaining a clearly defined meditation experience criterion. The second group comprised 12 novice participants, corresponding to subjects sub-013 through sub-024. Unlike the experienced practitioners, these individuals were familiar with meditation but did not maintain the same level of regular or intensive practice. Their limited meditation experience provided a contrasting participant category for investigating differences in EEG characteristics associated with meditation expertise. The mean age of the novice group was reported as 45.0 ± 14.8 years.

The dataset includes 24 participants, comprising 12 experienced meditators and 12 novice participants, ensuring an equal representation of the two categories at the participant level. However, after EEG segmentation and epoch extraction, the resulting dataset contains 8,751 epochs from experienced meditators and 4,318 epochs from novice participants. Thus, the epoch-level distribution is imbalanced, with approximately 67% of the epochs belonging to experienced meditators and 33% to novice participants. Therefore, equal participant representation should not be interpreted as class balance at the epoch level. This imbalance has been considered while evaluating the classification performance using class-wise precision, recall, and F1-score, along with balanced accuracy. The performance is also compared against a majority-class baseline to assess whether the classifier provides meaningful discrimination beyond simply predicting the dominant class.Consequently, the classifier can be trained and evaluated without requiring class-weight adjustments solely to compensate for unequal group sizes. The distinction between experienced and novice participants also provides a meaningful basis for examining whether EEG-derived time-frequency patterns contain sufficient information to differentiate individuals according to their meditation experience.

For the present study, the dataset serves as the common source of EEG recordings for both the conventional machine learning and deep learning approaches implemented within the NeuroState framework. Maintaining a common dataset and consistent preprocessing procedure allows the subsequent classifiers to be compared under equivalent experimental conditions. The multichannel nature of the recordings, together with their high original sampling frequency and clearly defined participant groups, makes ds001787 particularly suitable for investigating time-frequency EEG representations and evaluating automated approaches for meditation-related cognitive state classification [10].

3.2 Preprocessing Pipeline

A standardized EEG preprocessing procedure was implemented using the MNE-Python framework [1] to improve signal quality and obtain a consistent representation across participants. The preprocessing sequence was designed to reduce unwanted frequency components, minimize common-mode interference, identify contaminated signal segments, and generate fixed-duration EEG samples for subsequent time-frequency analysis. The complete procedure consisted of four major stages, as described below.

Stage 1: Channel mapping and electrode selection. The raw recordings obtained from the BioSemi acquisition system contained multiple channels identified using the amplifier-specific labeling convention. The channels were first mapped to conventional EEG electrode positions using the standard 10–20 electrode nomenclature. This step ensured that the spatial identity of each recording channel was consistently represented during subsequent processing. From the available 64-channel EEG montage, a subset of 19 electrodes was selected for analysis. The retained electrodes were FP1, FP2, F7, F3, FZ, F4, F8, T7, C3, CZ, C4, T8, P7, P3, PZ, P4, P8, O1, and O2. These electrodes provided coverage across the major cortical regions, including the frontal, temporal, central, parietal, and occipital areas. The selection was also consistent with the electrode configuration adopted in the previous investigation [2], thereby providing methodological continuity with the earlier work.

Stage 2: Frequency-domain filtering. The EEG recordings were subsequently subjected to frequency filtering to remove unwanted low-frequency drift and high-frequency interference. A linear-phase finite impulse response bandpass filter with cutoff frequencies of 4 Hz and 45 Hz was applied to each selected EEG channel. The lower cutoff attenuated slow baseline fluctuations and signal drift, whereas the upper cutoff reduced unwanted high-frequency components that were outside the principal frequency range considered in the present analysis. This filtering range retained the major EEG rhythms relevant to cognitive and meditative state analysis, including theta and alpha activity. In addition to bandpass filtering, a 50-Hz notch filter was applied to suppress interference originating from the electrical power supply. The combined filtering operation therefore produced cleaner EEG recordings while preserving the frequency information required for subsequent feature extraction and classification.

Stage 3: Average reference transformation. After spectral filtering, the EEG data were converted to an average reference. In this procedure, the instantaneous mean potential calculated across the available EEG channels was subtracted from the signal recorded at each electrode. This transformation reduces activity that is common across a large portion of the scalp and can consequently help minimize common-mode contributions. It also establishes a consistent reference condition across recordings, which is important when comparing EEG patterns between participants and across experimental sessions.

Stage 4: Epoch generation and artifact screening. Following filtering and rereferencing, the continuous EEG recordings were segmented into fixed-duration samples. Specifically, the data were divided into non-overlapping 5-second epochs using fixed-length event markers. Segmenting the continuous recordings into uniform intervals provided standardized input samples for subsequent wavelet decomposition and machine learning analysis. An amplitude-based artifact rejection criterion was then applied to identify epochs containing substantial contamination. An epoch was excluded whenever the peak-to-peak amplitude exceeded 100 μV in at least one channel. This criterion was applied at the epoch level so that segments containing pronounced artifacts, such as large transient disturbances, could be removed before model training. The rejection process was intentionally applied consistently

to all participants to maintain a uniform quality control procedure.

The number of usable epochs varied considerably between participants, ranging from approximately 15 to 1,079 retained segments per subject. Such variation is expected in EEG studies because recording duration, physiological artifacts, electrode contact, and individual signal quality can differ substantially across participants. After completion of the artifact rejection stage, a total of 13,069 clean 5-second EEG epochs were retained from the complete dataset. These artifact-reduced segments formed the input to the subsequent time-frequency transformation and classification stages of the proposed NeuroState framework.

3.3 Feature Extraction via Morlet Wavelet Decomposition

Time-frequency power maps were computed through Morlet wavelet convolution, implemented using the tfr_morlet routine in MNE-Python [1]. The Morlet wavelet family is generated from a Gaussian windowed complex exponential, whose mother wavelet takes the following form:

$$\psi(t) = e^{i2\pi ft}\, e^{-\frac{t^2}{2\sigma_t^2}} \tag{1}$$

where, f is the centre frequency, t is time, and σ_t is the temporal standard deviation of the Gaussian envelope.

$$W_x(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} x(t)\, \psi^* \left(\frac{t-b}{a} \right) \mathrm{d}t \tag{2}$$

where, b is the temporal translation, a is the scale parameter that controls center frequency, The Gaussian envelope in the frequency domain minimizes the Heisenberg time-frequency uncertainty product, and convolution is computed efficiently via the fast Fourier transform.

For the present study, decomposition was restricted to the theta band using four linearly spaced center frequencies: 4, 5, 6, and 7 Hz. The cycle count for each frequency was set to freqs/2, trading some frequency precision for improved temporal localization at lower frequencies. A decimation factor of 4 was applied along the time axis during computation, reducing effective temporal sampling from 256 Hz to 64 Hz and substantially lowering memory overhead.

The output of the decomposition was a four-dimensional array of instantaneous power values with dimensions [epochs $\times$ channels $\times$ frequencies $\times$ time-samples]. Averaging across the time axis collapsed this to [epochs $\times$ channels $\times$ frequencies], yielding one compact power feature tensor per epoch. For the deep learning pipeline, each tensor was treated as a 19 $\times$ 4 spatial map and extended with a singleton feature channel dimension prior to convolution. For classical models, the same tensor was flattened to a 76-dimensional vector (19 channels $\times$ 4 frequencies), standardized, and compressed by PCA [8].

3.4 Classical Machine Learning Pipeline

Feature vectors were first standardized by removing the mean of the training set and dividing by its standard deviation (zero-mean, unit-variance normalization). PCA [8] was then applied to the standardized training features, retaining the minimum number of components required to account for 95% of cumulative variance. Both the standardization parameters and the PCA projection matrix were estimated on the training partition and applied without re-fitting to the test partition, ensuring no information leakage. The resulting low-dimensional vectors trained a random forest [4] (100 estimators, Gini impurity splitting) and a support vector machine [3] with a radial basis function kernel (regularization parameter C = 1.0).

3.5 Wide Squeeze-and-Excitation Residual Network Architecture

The deep learning classifier is a wide squeeze-and-excitation residual network whose distinguishing features relative to a standard residual network are: an enlarged initial filter count (64 rather than the conventional 16–32), progressive filter doubling across three residual stages, and the insertion of squeeze-and-excitation attention blocks after each residual pair. The architecture processes input tensors of shape [19 $\times$ 4 $\times$ 1] and operates as follows [9, 10]:

(i) Input regularization. Gaussian noise with standard deviation σ = 0.02 is injected during training to discourage excessive reliance on precise amplitude values and improve robustness to electrode impedance fluctuations.

(ii) Stem block. A 3$\times$3 convolutional layer with 64 filters, followed by batch normalization and Swish activation, projects the raw input into a high-dimensional feature space from the outset.

(iii) Squeeze-and-excitation residual stages. Three stacked residual blocks with filter depths of 64, 128, and 256 learn progressively abstract representations. Within each block, two successive 3$\times$3 convolutions (each followed by batch normalization and Swish) are succeeded by a squeeze-and-excitation module: global average pooling collapses the spatial dimensions to a channel descriptor, two fully connected layers with reduction ratio r = 16 produce a soft gate vector via sigmoid activation, and this gate is applied to the feature maps by channel-wise multiplication. Projection shortcuts (1$\times$1 convolution and batch normalization) align dimensions across stages. Dropout (p = 0.2

and p = 0.3 at the first and subsequent stages) and Pooling window: (2*2), Stride: (2*2) alongside L2 weight decay λ = 0.0001 on all convolutional layers.

(iv) Classification head. Global average pooling converts the final spatial feature maps to a 256-dimensional vector. A fully connected layer with 128 units (Swish activation, p = 0.4 dropout, L2 regularization) precedes a sigmoid output unit that produces the expert/novice posterior probability. Optimization used the Adam algorithm with an initial learning rate of 0.001 and binary cross-entropy loss incorporating label smoothing (ε = 0.1) to reduce the risk of overconfident probability estimates. A ReduceLROnPlateau scheduler (a decay factor of 0.2 and a patience of four epochs) was paired with early stopping, which monitored validation accuracy with a patience of 12 epochs and restored the best-performing model weights. Training was conducted for at most 60 epochs with a batch size of 64; 20% of the training partition was reserved as a validation set. Each residual block comprises two successive 3×3 convolutional layers, with batch normalization and Swish activation applied as specified in the implementation. The output of the second convolutional layer was passed through a squeeze-and-excitation module before being combined with the residual shortcut. The squeeze-and-excitation module performed global average pooling to obtain channel-wise descriptors, followed by two fully connected layers with a reduction ratio of 16 and sigmoid activation to generate channel recalibration weights. These weights were multiplied with the convolutional feature maps to emphasize informative features and attenuate less relevant responses. Where the input and output dimensions differed between residual stages, a 1×1 projection shortcut with batch normalization was used to align their dimensions before residual addition [11–14].

4 Results and Discussion

4.1 Data Composition and Train-Test Split

After preprocessing and artifact rejection, 13,069 clean EEG epochs were retained from the 24 participants, comprising 8,751 epochs from experienced meditators and 4,318 epochs from novice meditators. The dataset was divided into training and test sets using a stratified 67/33 epoch-level split, resulting in 10,455 training epochs and 2,614 test The random seed used for partitioning was 42.

Importantly, the split was performed at the epoch level rather than at the subject level. Consequently, epochs originating from the same participant could be present in both the training and test sets. Since EEG recordings from an individual may exhibit subject-specific characteristics, this partitioning strategy may introduce dependence between the training and test samples and potentially result in an optimistic estimate of generalization performance to previously unseen individuals. Accordingly, the reported classification results should be interpreted as performance under an epoch-level evaluation protocol rather than as definitive evidence of subject-independent classification capability.

4.2 Training Behavior of the Wide Squeeze-and-Excitation Residual Network

The wide squeeze-and-excitation residual network converged over 54 epochs before early stopping intervened, ultimately recovering the weights from epoch 44—the point of peak validation accuracy (98.13%). Training accuracy at that checkpoint stood at 99.72%. The ReduceLROnPlateau scheduler triggered four reductions ($1 \times 10^{-3} \rightarrow 2 \times 10^{-4} \rightarrow 4 \times 10^{-5} \rightarrow 8 \times 10^{-6} \rightarrow 1.6 \times 10^{-6}$), each yielding a brief accuracy plateau followed by renewed improvement—a pattern diagnostic of learning-rate-sensitive convergence on a problem whose loss landscape contains several flat regions. Crucially, training and validation accuracy remained closely aligned throughout training, indicating that the combination of label smoothing, dropout, weight decay, and Gaussian input noise provided sufficient regularization for the dataset size.

4.3 Quantitative Performance Comparison

Table 1 presents the principal performance metrics for all three classifiers evaluated on the held-out test set. The wide squeeze-and-excitation residual network attained an accuracy of 98.29%, corresponding to precision and recall values of 0.98 and 0.99, respectively, for the binary task. The support vector machine and the random forest achieved accuracies of 88.59% and 81.65%, with the deep learning model outperforming these conventional classifiers by 9.7 and 16.6 percentage points, respectively.

Table 1. Performance comparison on the OpenNeuro ds001787 test set

Model	Accuracy	Precision	Recall	F1-Score
Wide squeeze-and-excitation residual network (proposed)	98.27%	98.77%	99.01%	98.89%
Support vector machine [3]	88.59%	72%	70%	71%
Random forest [4]	81.65%	78%	75%	76%

Examining the per-class breakdown via the confusion matrices provides additional nuance, as shown in Table 2.

Table 2. Confusion matrix of the proposed wide squeeze-and-excitation residual network on the test set

Actual/Predicted	Experienced	Novice
Experienced	True positive = 2,000	False negative = 20
Novice	False positive = 25	True negative = 569

These figures are interpretable through the lens of information preservation. The squeeze-and-excitation residual network receives the complete 19 × 4 power map as a structured grid, allowing its convolutional filters to detect coactivation patterns across neighboring channels and frequency bins—precisely the kind of distributed spatial pattern that meditation research consistently identifies as the neural substrate of meditative states as shown in Table 3. The classical pipelines, by contrast, first flatten this grid to a 76-dimensional vector and then further compress it via PCA; although 95% of variance is retained, the structural topology is irrecoverably lost. As a consequence, the random forest and the support vector machine are blind to correlations that are visible only when channels and frequencies are treated as a map rather than as a list.

Table 3. Architecture of the proposed wide squeeze-and-excitation residual network

Network Stage	Convolutional Layers	Output Dimensions	Description
Input	—	19 × 4 × 1	EEG channel–frequency power map
Stem	1	19 × 4 × 64	3 × 3 convolution; batch normalization; Swish activation
Residual stage 1	2	19 × 4 × 64	Two 3 × 3 convolutions; squeeze-and-excitation module; dropout rate of 0.2
Residual stage 2	2	9 × 2 × 128	Two 3 × 3 convolutions; squeeze-and-excitation module; dropout rate of 0.3
Residual stage 3	2	4 × 1 × 256	Two 3 × 3 convolutions; squeeze-and-excitation module; dropout rate of 0.3
Global average pooling	—	256	Spatial feature aggregation
Fully connected layer	—	128	Swish activation; dropout rate of 0.4; L2 regularization
Output layer	—	1	Sigmoid activation; experienced/novice classification

Note: EEG = electroencephalography; — = not applicable.

A second contributing factor is the squeeze-and-excitation attention mechanism itself. By learning per-channel scaling weights through global average pooling and a two-layer bottleneck, the squeeze-and-excitation module can suppress electrode locations that contribute principally noise—a property that is particularly valuable in EEG given the wellknown susceptibility of peripheral electrodes to muscle artifacts that survive threshold-based epoch rejection. The progressive widening of filter counts (64 → 128 → 256) ensures that increasingly abstract and combination-sensitive feature detectors are available at deeper stages.

Comparing these results to related work is instructive. The classification performance reported in this study should be interpreted in the context of its specific experimental objective. Unlike the present work [2], which investigates multiclass EEG classification involving control states, and it focuses on binary discrimination between experienced meditators and novice participants. Differences in class composition, EEG recordings, preprocessing procedures, and data-partitioning strategies may affect the reported classification accuracies. Consequently, the results of these studies represent different experimental settings and are not directly comparable. The proposed framework is evaluated primarily based on its ability to distinguish meditation expertise levels within the dataset considered in this study. Overall, these results make a clear empirical case: preserving the structural information contained in two-dimensional time-frequency representations is important for EEG-based meditation classification, and deep architectures that preserve and exploit that structure substantially outperform pipelines that reduce it to unordered feature vectors before classification.

4.4 Evaluation Protocol and Generalization Limitations

The performance reported in this study was obtained using an epoch-level train–test split. Because multiple epochs from the same participant may occur in both partitions, the possibility of subject-specific information influencing the classification results cannot be excluded. This limitation is particularly relevant to EEG analysis, where individual differences in signal characteristics may contribute to classification performance, Although the proposed wide squeeze-and-excitation residual network achieved a test accuracy of 98.29% under the adopted evaluation protocol, this result should not be interpreted as establishing equivalent performance on entirely unseen participants. Future evaluation should employ subject-independent partitioning, such as GroupShuffleSplit or StratifiedGroupKFold, with participant identity used to ensure that epochs from a given subject do not appear in both training and test sets. Such experiments would provide a more rigorous assessment of the model's ability to generalize across individuals.

The performance differences among the three classifiers may be attributed to their distinct feature representation and learning strategies. In the conventional machine learning pipeline, PCA reduces the dimensionality of the extracted features before classification using the support vector machine or the random forest. Although PCA can reduce redundancy and computational complexity, it transforms the original feature space into a lower-dimensional representation that may not preserve all discriminative information relevant to distinguishing experienced meditators from novices. Furthermore, the support vector machine and the random forest operate on the resulting feature vectors and do not explicitly exploit the spatial relationships between EEG channels and frequency components.

In contrast, the proposed wide squeeze-and-excitation residual network learns hierarchical representations directly from the two-dimensional EEG power maps. This approach allows convolutional layers to capture local patterns and relationships among channels and frequency components, while residual connections facilitate feature learning across multiple network stages. The squeeze-and-excitation mechanism may further improve representation learning by adaptively recalibrating feature channel responses according to their global information. Consequently, the network can emphasize potentially informative feature maps and suppress less useful responses. These architectural properties provide a plausible explanation for the higher classification accuracy observed with the wide squeeze-and-excitation residual network. However, the individual contributions of the wider residual architecture and squeeze-and-excitation mechanism require ablation experiments to be established conclusively.

The observed results are also consistent with the broader direction of EEG classification research, in which deep-learning approaches are investigated for their ability to learn task-relevant representations from structured EEG data, whereas conventional methods often depend on predefined features and dimensionality reduction techniques. In the present study, the wide squeeze-and-excitation residual network achieved an accuracy of 98.29%, compared with 88.59% for the support vector machine and 81.65% for the random forest. These results suggest that learning directly from two-dimensional EEG power representations may be advantageous for the present meditation classification task. Nevertheless, comparisons with previous studies should account for differences in datasets, participant populations, preprocessing procedures, validation strategies, and evaluation metrics. In particular, because the current evaluation uses an epoch-level data split, the reported performance should be interpreted cautiously until it is validated using a participant-independent evaluation protocol.

5 Conclusion

This study investigated the effectiveness of preserving the two-dimensional structure of EEG power representations for meditative state classification using the OpenNeuro dataset. The proposed wide squeeze-and-excitation residual network achieved a reported test accuracy of 98.29%, outperforming the PCA-based support vector machine and random forest classifiers, which achieved accuracies of 88.59% and 81.65%, respectively. These findings suggest that learning directly from structured EEG power maps can provide an advantage over conventional feature-based classification approaches for the task considered in this study. The performance of the wide squeeze-and-excitation residual network may be attributed to the complementary roles of its architectural components. Residual connections facilitate the learning of hierarchical feature representations, while the squeeze-and-excitation mechanism adaptively recalibrates feature map responses to emphasize potentially informative patterns. Additionally, dropout and label smoothing help regularize the learning process and may reduce overfitting. Although these design choices offer plausible explanations for the observed performance, further ablation experiments are required to quantify the individual contribution of each component.

Overall, the results highlight the importance of EEG representation and model architecture in distinguishing experienced meditators from novices. Future work will investigate Transformer-based architectures for capturing broader relationships among EEG channels and multimodal approaches that integrate EEG with complementary physiological signals, such as heart rate measurements. Furthermore, participant-independent validation on larger and more diverse datasets will be essential to assess generalization to unseen individuals and establish the robustness of the proposed framework for practical meditative state monitoring applications. Further research may explore attention-based and hybrid deep learning architectures to improve classification accuracy, robustness, and generalization across subjects.

Author Contributions

Conceptualization, V.K. and A.P.S.; methodology, A.P.S. and M.S.; software, A.P.S. and M.S.; validation, A.P.S., M.S., and A.R.; formal analysis, A.P.S. and V.K.; investigation, A.P.S. and A.R.; resources, V.K.; data curation, A.P.S. and M.S.; writing—original draft preparation, A.P.S. and M.S.; writing—review and editing, V.K. and A.R.; visualization, M.S.; supervision, V.K.; project administration, V.K.; funding acquisition, V.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] A. Gramfort, M. Luessi, E. Larson, D. A. Engemann, D. Strohmeier, C. Brodbeck, R. Goj, M. Jas, T. Brooks, L. Parkkonen *et al.*, "MEG and EEG data analysis with MNE-Python," *Front. Neurosci.*, vol. 7, pp. 1–13, 2013. https://doi.org/10.3389/fnins.2013.00267

[2] S. Usgaonkar, D. R. Edla, and R. R. Reddy, "A meditation-based brain state classification framework: An integrated Morlet wavelet transforms and CNN approach with EEG signals," *Multimed. Tools Appl.*, vol. 84, pp. 47 911–47 941, 2025. https://doi.org/10.1007/s11042-025-21065-w

[3] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, 1995. https://doi.org/10.1007/BF00994018

[4] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001. https://doi.org/10.1023/A:1010933404324

[5] R. T. Schirrmeister, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggensperger, M. Tangermann, F. Hutter, W. Burgard, and T. Ball, "Deep learning with convolutional neural networks for EEG decoding and visualization," *Hum. Brain Mapp.*, vol. 38, no. 11, pp. 5391–5420, 2017. https://doi.org/10.1002/hbm.23730

[6] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015. https://doi.org/10.1038/nature14539

[7] S. Mallat, *A Wavelet Tour of Signal Processing*. San Diego, CA, USA: Academic Press, 1999.

[8] I. T. Jolliffe, *Principal Component Analysis*. New York, NY, USA: Springer, 2002.

[9] A. Delorme and T. Brandmeyer, "EEG meditation study," OpenNeuro, 2018. https://doi.org/10.18112/openneuro.ds001787.v1.0.0

[10] X. Niu, N. Lu, R. Yan, and H. Luo, "A time–frequency map generation network embedded with spectral self-attention for motor imagery classification," *Biomed. Signal Process. Control*, vol. 93, p. 106206, 2024. https://doi.org/10.1016/j.bspc.2024.106206

[11] M. Miao, J. Liang, Z. Sheng, W. Liu, B. Xu, and W. Hu, "ST-SHAP: A hierarchical and explainable attention network for emotional EEG representation learning and decoding," *J. Neurosci. Methods*, vol. 414, p. 110317, 2025. https://doi.org/10.1016/j.jneumeth.2024.110317

[12] Y. Ni, Q. Cai, H. Wang, and Y. Wang, "TFS-FENet: A time-frequency spatial deep learning framework for EEG-based ADHD subtype classification," *Appl. Neuropsychol.: Child*, pp. 1–11, 2025. https://doi.org/10.1080/21622965.2025.2606108

[13] T. Bayan, D. Das, and N. Deb, "Deep hybrid CNN–BiLSTM–Attention model for EEG classification using wavelet features," *Ann. Neurosci.*, p. 09727531251396337, 2026. https://doi.org/10.1177/09727531251396337

[14] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 7132–7141. https://doi.org/10.1109/CVPR.2018.00745